INCH-POUND

MIL-S-71027 (AR)
02 MAY 1991

MILITARY SPECIFICATION

SIGHTUNIT: M67

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for one type of instrument known as Sightunit: M67 (9356182) used with U.S. Army mortar systems. Throughout this document, this instrument is referenced as "sightunit." This sightunit is composed of Telescope, Elbow: 9356181 and Mount, Telescope: 9356166. This sightunit has radioactively illuminated scales, level vials, indices, and reticle (see 3.1 and 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-T-71029	-	Telescope, Elbow: 9356181
MIL-M-71028	-	Mount, Telescope: 9356166

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1240

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9356166 - Mount, Telescope: 9356166
9356181 - Telescope, Elbow: 9356181
9356182 - Sightunit: M67
11750709 - Adapter, Final Inspection, Azimuth

SPECIAL PACKAGING INSTRUCTION

PS 9356182 - Packaging of Sightunit: M67

(Copies of drawings, publications, and other Government documents required by contractors in connection the specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DLAM 4145.8 - Radioactive Commodities in the DOD Supply System

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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DEPARTMENT OF TRANSPORTATION, BUREAU OF HAZARDOUS MATERIAL

49 CFR 171-190 - Interstate Commerce Commission and Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

CODE OF FEDERAL REGULATIONS

Title 10 (Nuclear Regulatory).

Part 20: Standards for Protection against radiation.

Part 30: Rules of General Applicability to Licensing of by-product material.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Radioactive material caution. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, possession, handling, and shipping of radioactive material.

3.2 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.3 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9356182, all associated drawings, and with all requirements specified in applicable specifications and standards. The following subassemblies of the sightunit shall have met all requirements of their respective drawings and specifications prior to their installation in the sightunit.

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Name	Drawing No.	Specification
Telescope, Elbow	9356181	MIL-T-71029
Mount, Telescope	9356166	MIL-M-71028

Where applicable, the sightunit shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.4 Telescope clamping mechanism. When locked in place by applying a force of 10-12 in-lbs to thc wingnut and 15 - 17 in-lbs to the locking lever, there shall be no rotation of the telescope within the inside diameter of the clamp when a 35 in-lb torque is applied and there shall be no pivoting of the clamping mechanism when 80 in-lb torque is applied. When unlocked, the telescope and clamping mechanism shall rotate freely.

3.5 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.6 Workmanship. The workmanship requirements of MIL-F-13926 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+60°F to +90°F (+16°C to 32°C)
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg. (700 to 815 mmHg)

4.3 First article inspection.

4.3.1 Sample. The contractor shall submit a first article sample, as designated by the contracting officer, for evaluation in accordance with provisions of 4.3.2. The sample consists of the subassemblies listed below in the quantity indicated.

Name	Drawing	Specification	Quantity
Mount, Telescope	9356166	Mil-M-71028	3
Telescope, Elbow	9356181	Mil-T-71029	3

4.3.2 Inspections to be performed. Each subassembly refered in 4.3.1 shall have met all the examinations and tests as specified in their detail specification prior to their installation in the sightunit. The first article sample shall be subjected to all of the examination specified in Table I (functional tests) herein. It may be inspected for compliance with any or all requirements of the applicable drawings or QAPs.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to

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determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package List (TDPL). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

a. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with DLAM 4145.8 and Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests
(100% Inspection)

Characteristic	Requirement	Test Procedure
Critical:	None	N/A
Major:		
101. Telescope clamping mechanism	3.4	4.7.1
Minor:	None	N/A

4.5 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed

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product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Inspection equipment design requirements.

4.6.2.3.1 Telescope clamping mechanism. The Adapter, Final Inspection, Azimuth, 11750709, together with a suitable stand, shall be utilized in performing the telescope clamping mechanism test. The stand shall have the necessary stability for positioning the adapter so that it maintains the mounting dovetail surface and dovetail axis of the mount in a vertical plane within ± one (1) degree.

4.6.2.3.2 Wall target. The wall target shall contain a vertical and horizontal crossline. It shall be positioned 6 to 9 feet in front of the sightunit under test.

4.7 Methods of inspection.

4.7.1 Telescope clamping mechanism. Conduct this test with the mount positioned on the testing equipment specified in 4.6.2.3.1. Position the target so that the intersecting point of the vertical and horizontal target lines coincide with the geometric axis of the clamp for the telescope. Install the telescope in the clamping mechanism of the mount and lock the clamp by applying a force of 15 to 17 in-lbs to the locking lever and 10 to 12 in-lbs to the wingnut. By means of the elevation and azimuth knobs, adjust the line of sight of the telescope so that its vertical and horizontal reticle lines are superimposed on the image at the vertical and horizontal target lines. The telescope clamping mechanism shall conform to the requirements of 3.4 when the applicable torques are applied to the telescope body and the clamp for the telescope. Any rotation of the telescope within the clamp, or pivoting of the clamping mechanism about its own axis shall be determined by the permanent displacement (lack of coincidence) of the telescope reticle lines with respect to the target image lines. If the requirements are not met, the item shall be rejected.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PS 9356182 and DSAM 4145.8. The level of protection shall be as specified in the procurement document.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Sightunit, M67, is used for laying the U.S. Army mortar systems in deflection and elevation for direct and indirect fire modes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation.
c. Applicable stock number.
d. Applicable packaging data sheet number (see 5.1).
e. Provisions for First Article Testing.
f. Certificate of conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Mortar
Mount, Telescope
Telescope, Elbow
Tritium

6.6 Issue. This is an initial release.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1240-AB32)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-71027 (AR)	2 May 1991

3. DOCUMENT TITLE

SIGHTUNIT: M67

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army, ARDEC Specifications and Standardization Office	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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